

Buffer module - design brief

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A replaceable purchased energy absorber, bumper face and bolted mounting plate for a later crane end-stop programme. Absorber internals and force-stroke characteristics are not designed.

TARGETS (not validated)

Study impact energy: 5 kJ. Illustrative requirements input

Illustrative stroke: 150 mm. Reserved travel

Average stopping force: 33.3 kN. E / s only; peak force unknown

Supplier rating: Unassigned. Impact mass, speed and duty required

DESIGN DECISIONS

Partner with an absorber specialist rather than infer impact qualification from brake work.

Use a replaceable mounting interface and visible stroke envelope.

Size the machine attachment for verified peak force, not E/s average.

CANDIDATE MATERIALS

Mounting plate: S355J2+N candidate.

Absorber and bumper face: supplier-controlled hydraulic/elastomer components.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Collect impact mass, speed, available stroke and recovery interval.
- 2. Select rated absorber and replace CAD envelope with supplier model.
- 3. Approve foundation fasteners and collision clearance.

OPEN DESIGN RISKS

- 1. No metering geometry, damping curve or peak force established.
- 2. Travel envelope does not demonstrate energy absorption.

VALIDATION REQUIRED

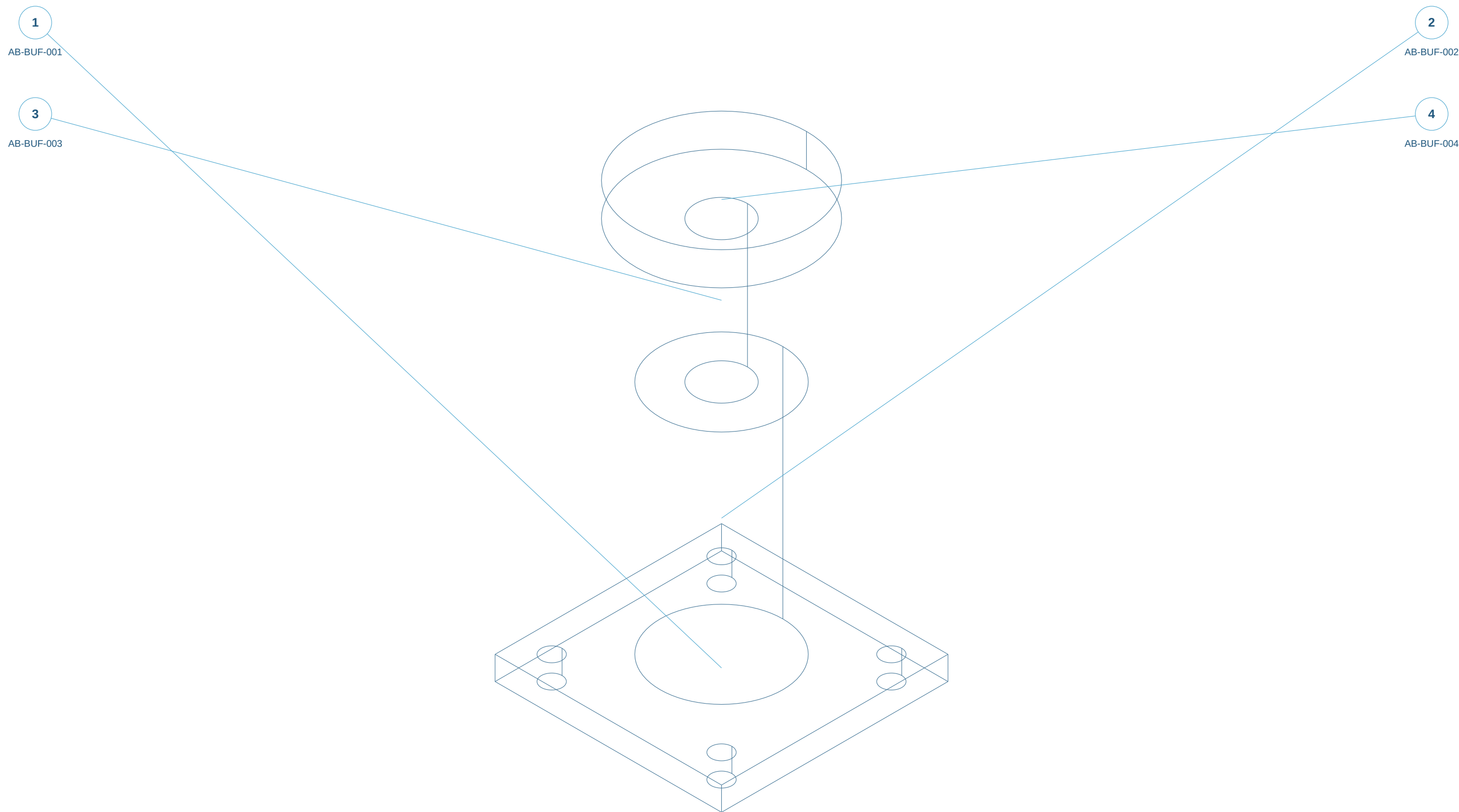
- 1. Supplier force-stroke curve and machine structure analysis.
- 2. Instrumented impact, temperature and repeated-impact tests.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Stroke/end-position and impact-event recording only; define survival environment and cable protection.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-BUF-001 | Mounting plate | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-BUF-002 | Absorber body envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy rated absorber

AB-BUF-003 | Rod envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Vendor stroke model required

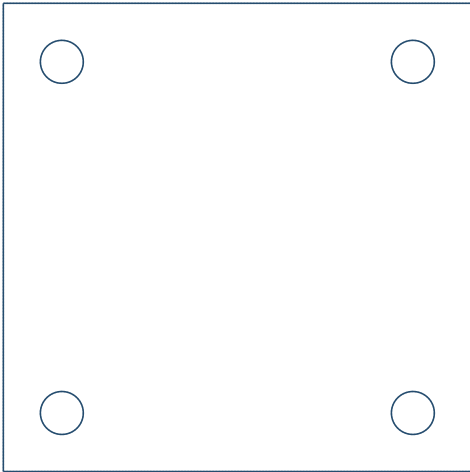
AB-BUF-004 | Impact face envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy supplier face/bumper

AB-BUF-001 / Mounting plate

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 240.00 / 240.00 / 25.00 mm

240 x 240 x 25; 4 x diameter 22 on 180 square.

Foundation bolts must be sized for supplier peak force.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

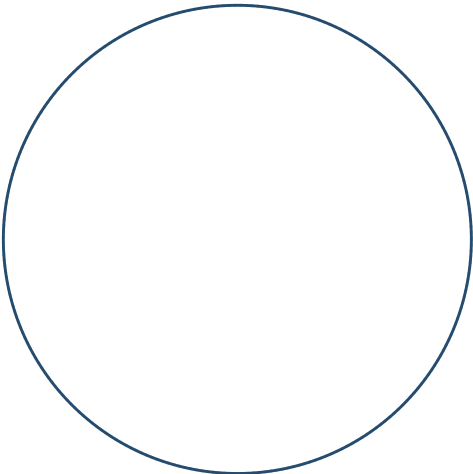
SIDE Y-Z



AB-BUF-002 / Absorber body envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy rated absorber

Overall X/Y/Z bounds: 130.00 / 130.00 / 250.00 mm

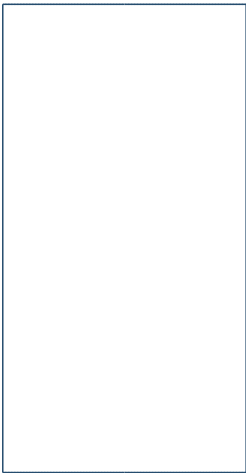
All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

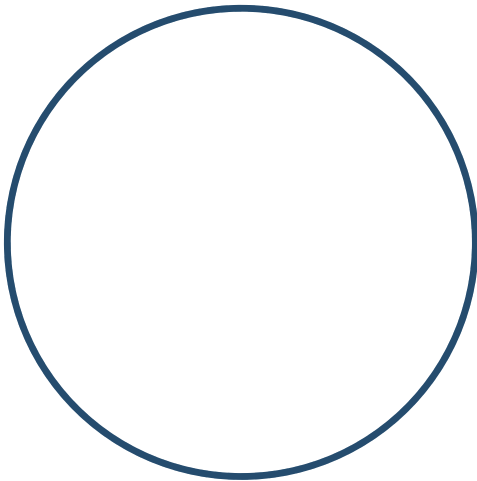
SIDE Y-Z



AB-BUF-003 / Rod envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Vendor stroke model required

Overall X/Y/Z bounds: 55.00 / 55.00 / 150.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

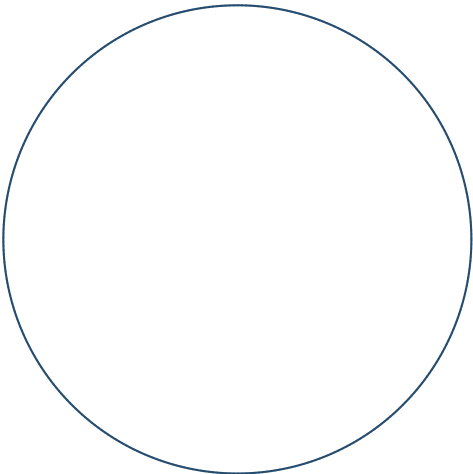
SIDE Y-Z



AB-BUF-004 / Impact face envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy supplier face/bumper

Overall X/Y/Z bounds: 180.00 / 180.00 / 35.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

